

[Aim: 100/100 in Maths]

आरंभ

PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

Lecture #8

#LP: Meena went to a bank to withdraw Rs. 2000. She asked the cashier to give her Rs. 50 and Rs. 100 notes only. Meena got 25 notes in all. Find how many Rs. 50 and Rs. 100 she received.

$$\text{Total amount} = \text{₹ } 2000$$

$$\text{₹ } (50x) + \text{₹ } (100y) = 2000$$

$$50x + 100y = 2000$$

$$\cancel{50(x + 2y) = 2000}$$

$$x + 2y = 40 \quad \text{--- (I)}$$



'x' notes



'y' notes

$$\text{total notes} = 25$$

$$x + y = 25 \quad \text{--- (2)}$$

$$x = \text{₹ } 50 \text{ notes} =$$

$$y = \text{₹ } 100 \text{ notes} =$$

Age Wale Sawaal:

#LP: The present age of the father is 3 years more than three times the age of the son. Three years hence, the father's age will be 10 years more than twice the age of the son. Determine their present age.

	Present	3 yrs later
F	$(3 + 3x)$	$3 + 3x + 3 \Rightarrow (3x + 6)$
S	x (let)	$(x + 3)$

At 3 yrs hence,

$$F' = 10 + 2S'$$
$$\Rightarrow (3x + 6) = 10 + 2(x + 3)$$
$$\Rightarrow 3x + 6 = 10 + 2x + 6$$
$$\Rightarrow 3x - 2x = 10$$
$$x = 10$$

Final Ans

Present age

$$F = 3 + 3(10) = 33 \text{ yrs}$$
$$S = x = 10 \text{ yrs}$$

#LP: A father's age is three times the sum of the ages of his two children.
After 5 years his age will be two times the sum of their ages. Find the
present age of the father. (CBSE 2019)

Let ^{present} ages of two children x & y .

	Present	After 5 yrs
1st	x	$x+5$
2nd	y	$y+5$
F	$\Rightarrow$ <u>$3(x+y)$</u>	$\rightarrow 3(x+y)+5$

Ans 5 yrs hence

$$F' = 2(x' + y')$$

$$\Rightarrow 3(x+y)+5 = 2(x+5+y+5)$$

$$\Rightarrow 3x+3y+5 = 2x+10+2y+10$$

$$\Rightarrow \frac{3x-2x+3y-2y}{1} = 20-5$$

$$\Rightarrow \boxed{x+y=15}$$

Final Ans F present age $\Rightarrow 3(x+y)$

$$\Rightarrow 3(15)$$

$$\Rightarrow \boxed{45 \text{ yrs}}$$

#LP: Three years ago, Rashmi was thrice as old as Nazma. Ten years later, Rashmi will be twice as old as Nazma. How old are Rashmi and Nazma now? [CBSE 2024]

	3 yrs ago	Present	10 yrs later
R	$(x-3)$	x (let)	$(x+10)$
N	$(y-3)$	y (let)	$(y+10)$

ATQ 3 yrs ago

$$(x-3) = 3(y-3) \quad \text{--- (i)}$$

ATQ 10 yrs later

$$(x+10) = 2(y+10) \quad \text{--- (ii)}$$

#LP: A is elder than B by 2 years. A's father, F, is twice as old as A, and B is twice as old as his sister S. If the age of the father and sister differ by 40 years, find the age of A.

$$A = B + 2$$

$$A = x + 2$$

Let age of B is x .

300 Final M $A = x + 2$

$$= 24 + 2$$

$$A = 26 \text{ yrs}$$

A7Q

$$F = 2A$$

$$F = 2(x + 2)$$

A7Q

$$B = 2S$$

$$x = 2S$$

$$\frac{x}{2} = S$$

A7Q

$$F - S = 40$$

$$2(x + 2) - \frac{x}{2} = 40$$

$$\Rightarrow \frac{4(x + 2) - x}{2} = 40$$

$$4x + 8 - x = 80$$

$$3x = 80 - 8$$

$$3x = 72$$

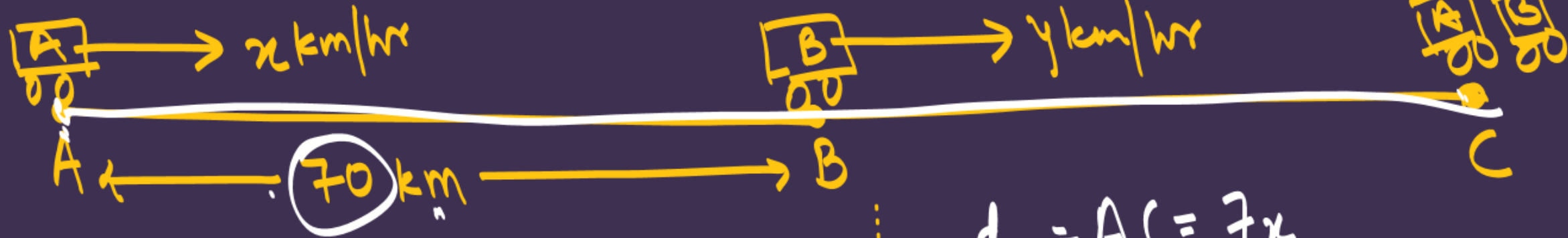
$$x = 24$$

Speed, Distance and Time:

$$D = S \times t$$

#LP: Points A and B are 70km. apart on a highway. A car starts from A, and another car starts from B simultaneously. If they travel in the same direction, they meet in 7hrs. ~~but if they travel towards each other they meet in 1 hr.~~ Find the speed of two cars.

Case 1



(A) $\Rightarrow$

$$t_A = 7 \text{ hrs}$$
$$S_A = x \text{ km/hr.}$$

$$d_A = S_A \cdot t_A$$

$$d_A = x(7)$$

$$d_A = 7x$$

(B)

$$t_B = 7 \text{ hr}$$
$$S_B = y \text{ km/hr.}$$

$$d_B = S_B t_B$$

$$d_B = y(7)$$

$$d_B = 7y$$

$$d_A = AC = 7x$$

$$d_B = BC = 7y$$

We can see $\Rightarrow AC - BC = 70 \text{ km}$

$$7x - 7y = 70$$

$$x - y = 10$$

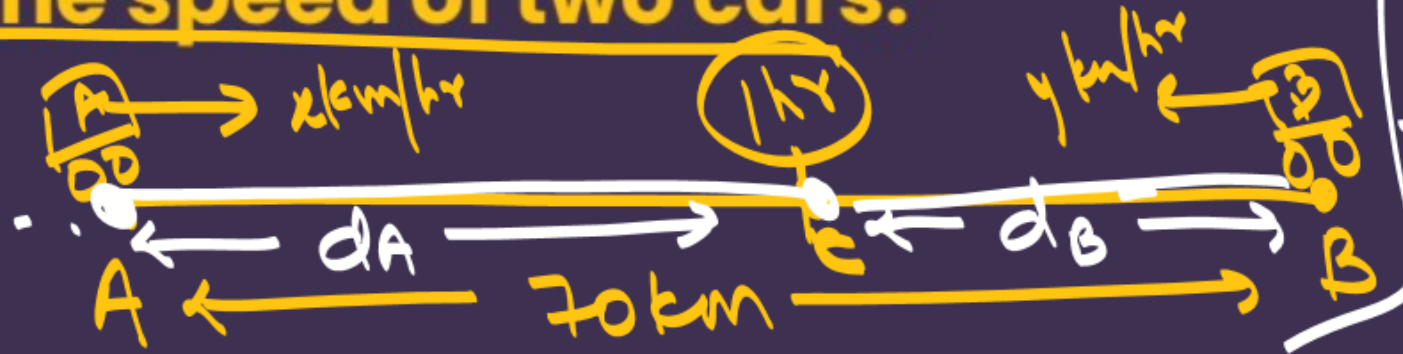
$$x - y = 10 \quad (1)$$

Speed, Distance and Time:

$$D = S \times t$$

#LP: Points A and B are 70km. apart on a highway. A car starts from A, and another car starts from B simultaneously. ~~If they travel in the same direction,~~ but if they travel towards each other, they meet in one hour. Find the speed of two cars.

Case II



2 cars

$$AC + BC = 70$$
$$d_A + d_B = 70$$

$$x + y = 70 \quad \text{--- (I)}$$

(A) $t_A = 1 \text{ hr}$
 $S_A = x \text{ km/hr}$

$$d_A = S_A \cdot t_A$$
$$d_A = x(1)$$
$$d_A = x$$

(B) $t_B = 1 \text{ hr}$
 $S_B = y \text{ km/hr}$

$$d_B = S_B \cdot t_B$$
$$d_B = y(1)$$
$$d_B = y$$

$$\textcircled{2} \rightarrow x - y = 10$$

$$\textcircled{1} \rightarrow x + y = 70$$

$\oplus$

$$2x = 80$$

$$x = 40$$



$$40 + y = 70$$

$$y = 70 - 40$$

$$y = 30$$

Final Ans

$$S_A = x \text{ km/hr} = 40 \text{ km/hr}$$

$$S_B = y \text{ km/hr} = 30 \text{ km/hr}$$

#LP: Two people are 16 km apart on a straight road. They start walking at the same time. If they walk towards each other with different speeds, they will meet in 2 hours. Had they walked in the ~~same direction~~ with ~~the same speeds as before~~, they would have met in ~~2 hours~~. Find their walking speeds. [CBSE 2023]



(A) $t_A = 2 \text{ hr}$
 $S_A = x \text{ km/hr}$

$d_A = 2x$

(B) $t_B = 2 \text{ hr}$
 $S_B = y \text{ km/hr}$

$d_B = 2y$

we can see

$$AC + BC = 16 \text{ km}$$

$$d_A + d_B = 16$$

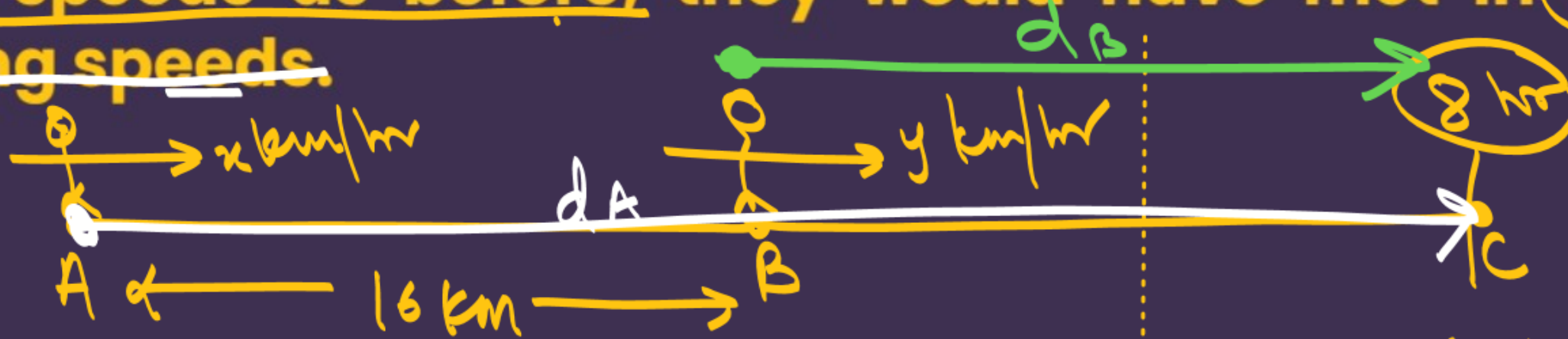
$$2x + 2y = 16$$

$$\cancel{x}(x+y) = \cancel{16}8$$

$$\boxed{x+y=8} \quad \text{--- (2)}$$

#LP: Two people are 16 km apart on a straight road. They start walking at the same time. If they walk towards each other with different speeds, they will meet in 2 hours. Had they walked in the same direction with same speeds as before, they would have met in 8 hours. Find their walking speeds. [CBSE 2023]

Ques II



(A) $t_A = 8 \text{ hr}$
 $S_A = x \text{ km/hr}$

$d_A = 8x$

(B) $t_B = 8 \text{ hr}$
 $S_B = y \text{ km/hr}$

$d_B = 8y$

We can see a $AC - BC = 16$

$d_A - d_B = 16$

$8x - 8y = 16$

$8(x - y) = 16$

$x - y = 2$ (I)

$$\textcircled{I} \rightarrow x + y = 8$$

$$\textcircled{II} \rightarrow x - y = 2$$

$\oplus$

$$\cancel{2x} = \cancel{10} \bar{5}$$

S_A

$$\Rightarrow x = 5 \text{ km/hr}$$

put

$\Downarrow$

$$5 - y = 2$$

$$5 - 2 = y$$

$\Downarrow$

$$\Rightarrow y = 3 \text{ km/hr}$$

final Ans

PLEQV → complete Module
Solve

THANK YOU COODIESS !!

Next week → Summer vacation

classes resume date → Weekly planner

